

Volatility Forecasting Using Spiking Neural Networks with Options-Derived Input Signals

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Abstract

Spiking Neural Networks (SNNs) provide an event-based framework for modelling temporal data, but their application to financial time series remains limited. After background research, it was found that minimal prior work has examined volatility forecasting using SNNs. This project will therefore aim to utilise the powerful algorithms of EventProp and e-prop to solve the forecasting problem of Variance Risk Premium (VRP) prediction. We aim to forecast the VRP, defined as the difference between implied and realised volatility, at weekly and monthly horizons. A secondary target, the relative VRP ($rVRP = (VIX - RV) / VIX$), is evaluated alongside the absolute VRP to assess whether level-normalised forecasts generalise better across volatility regimes.

Volatility is selected as the target because it exhibits persistence, clustering, and mean reversion, in contrast to asset returns which are difficult to predict under the Efficient Market Hypothesis. Weekly and monthly horizons are chosen because the dominant econometric benchmark, HAR-RV, exploits daily autocorrelation structure that weakens at longer horizons, giving the SNN implementation a more realistic opportunity to demonstrate value. The VRP is time-varying and has been shown to contain predictable structure. Input features are derived from options markets, including VIX, VVIX, SKEW, and the volatility term structure, which provide forward-looking information about expected volatility that is not available from price history alone.

The project will evaluate four spike encoding strategies, with a focus on delta or event-based encoding of volatility increments using adaptive per-series thresholds. The model is implemented in the GeNN/mGeNN framework and will incorporate loss shaping and heterogeneous neuron timescales to improve training stability. Performance is compared against GARCH, HAR-RV, and LSTM baselines at both time horizons. If the SNN matches or exceeds HAR-RV at weekly or monthly frequency, this would constitute the first demonstration of a neuromorphic architecture competitive with the established econometric standard for volatility forecasting and would support the case for options-derived signals as inputs to spiking models.

Introduction

Spiking Neural Networks (SNNs) encode information using discrete spike events rather than continuous activations. This makes them suitable for modelling temporal processes where information is carried by changes over time. While SNNs have been studied extensively in computational neuroscience and neuromorphic computing, their use in financial time series modelling is limited. Existing SNN finance work focuses on high-frequency price spike prediction (Gao et al., 2021; Ezinwoke and Amos, 2025), where the event-driven structure of

the data maps directly onto the SNN's computational paradigm. The application of SNNs to lower-frequency statistical forecasting tasks, where the target is a slowly varying quantity rather than a discrete event, remains unexplored.

Within financial markets, financial prediction is typically constrained by the Efficient Market Hypothesis, which implies that asset prices incorporate available information quickly. As a result, return prediction is difficult and often unstable. Volatility, however, exhibits persistent statistical structure, including clustering and mean reversion (Bollerslev, 1986; Andersen et al., 2003; Corsi, 2009). This makes volatility forecasting a more tractable problem.

The risk variance risk premium (VRP) is defined as the difference between implied volatility and realised volatility. The VRP is typically positive on average, reflecting compensation investors demand for bearing volatility risk. It varies over time and has been shown to be predictable using both econometric and machine learning methods (Carr and Wu, 2009; Bollerslev et al., 2009). This project forecasts the VRP at weekly and monthly horizons. These horizons are chosen deliberately: HAR-RV, the dominant econometric benchmark, exploits daily autocorrelation structure in realised variance across daily, weekly, and monthly components (Corsi, 2009). This advantage decays at longer horizons, reducing the benchmark's edge and giving the SNN a more realistic opportunity to demonstrate value. A secondary target, the relative VRP ($rVRP = (VIX - RV) / VIX$), is also evaluated. Unlike the absolute VRP, which varies in magnitude with the volatility level, the relative VRP is approximately scale-invariant. The model will be trained on pre-2020 data and tested on post-2020 data, which includes materially higher volatility regimes. Therefore, the rVRP reduces the distributional shift between training and test periods by controlling for the level effect.

Options markets provide information about expected future volatility. Implied volatility measures and related indices such as VIX, VVIX, and SKEW capture different aspects of the volatility surface, including level, uncertainty, and tail risk. These measures contain information about future realised volatility beyond that available from historical price data alone (Jiang and Tian, 2005; Atkins et al., 2018).

Existing applications of SNNs in finance focus mainly on high-frequency price prediction and trading strategies (Gao et al., 2021; Ezinwoke and Amos, 2025), as well as broader modelling tasks such as portfolio optimisation (Mohan et al., 2025). To the best of our knowledge, SNNs have not been applied to volatility forecasting, and no work has incorporated options-derived signals as inputs to an SNN model.

This project investigates whether SNNs trained using EventProp and/or e-prop can forecast the VRP and rVRP at weekly and monthly horizons. A central component is the use of spike-based encoding schemes, particularly delta or event-based encoding of volatility increments with adaptive thresholds. The project evaluates model performance against standard benchmarks and examines whether options-derived inputs improve forecast accuracy relative to price-based features alone.

Literature Review and Background

Volatility Forecasting:

Volatility forecasting has a well-established literature. GARCH (Bollerslev, 1986) models volatility as an autoregressive conditional process and remains the standard short-horizon benchmark. The HAR-RV model (Corsi, 2009) extends this by constructing a linear model from daily, weekly, and monthly components of realised variance, exploiting the heterogeneous autoregressive structure of realised volatility across market participants with different trading horizons. HAR-RV is widely regarded as the strongest econometric baseline at daily to monthly horizons; its advantage weakens at weekly and monthly horizons as the daily autocorrelation structure it relies on begins to decay. Deep learning methods including LSTMs have shown modest improvements over HAR-RV at short horizons, but gains are less consistent at longer horizons. Atkins et al. (2018) demonstrated that implied volatility and

news-derived signals improve forecast accuracy over price history alone. Carr and Wu (2009) and Bollerslev et al. (2009) established that the VRP contains predictable time-series structure and is forecastable using both reduced-form and structural models, motivating its use as the primary target variable here.

SNNs in Financial Applications:

Gao et al. (2021) introduced unsupervised spike learning for high-frequency trading, predicting price spikes in time series data and developing three backtested HFT strategies. This was the first demonstration of SNNs applied to financial data in a trading context and established the basic framework for encoding financial signals as spike trains. Ezinwoke and Amos (2025) extended this work, evaluating three SNN architectures on high-frequency AAPL price data with Bayesian-optimised hyperparameters. They introduced a novel Penalised Spike Accuracy (PSA) objective to align predicted spike rates with empirical rates, finding that unsupervised STDP-trained models significantly outperformed the supervised backpropagation baseline in backtesting, with the extended STDP model achieving a cumulative return of 76.8% against 42.54% for the supervised alternative.

Kasabov et al. (2023) applied evolving SNNs to multimodal financial streaming data including price series and news, demonstrating that spiking connectivity patterns provide interpretable insights into variable interactions over time. Matenczuk et al. (2021) compared traditional and spiking networks on ETF price forecasting and found comparable accuracy with reduced computational cost. Mohan et al. (2025) applied SNNs to cross-market portfolio optimisation, with the notable finding that SNN neuron firing density spontaneously correlated with market volatility periods without being trained to do so. No published work targets volatility forecasting directly with SNNs, and none incorporates options-derived signals as SNN inputs.

SNN Training Algorithms:

Bellec et al. (2020) introduced e-prop, an online learning rule approximating gradients through eligibility traces, avoiding full backpropagation through time and making it suited to streaming applications. Wunderlich and Pehle (2021) introduced EventProp, which computes exact gradients via the adjoint method. Nowotny et al. (2025) identified the spike deletion problem in EventProp: gradient descent can inadvertently silence useful hidden neurons because the exact gradient provides no signal when a synaptic weight crosses the threshold at which a spike disappears. The proposed solution: loss shaping via an exponentially weighted integrated membrane potential loss, is directly applicable to the regression setting of this project. They also demonstrated that heterogeneous neuron timescales, drawn from a gamma distribution, improve accuracy by allowing neurons to specialise across different temporal scales — a property with a direct financial interpretation when applied to volatility dynamics at multiple horizons.

Research Design and Methodology

Data:

Realised volatility will be computed from daily index return data using rolling variance estimators and aggregated to weekly and monthly frequencies. The Oxford-Man Institute Realised Library provides pre-computed daily estimates for major indices at no cost. Options-derived signals will be sourced from CBOE free daily publications:

- VIX: 30-day S&P 500 implied volatility
- VIX9D, VIX3M, VIX6M: short and medium-term implied vol for term structure
- VVIX: implied volatility of VIX options
- SKEW Index: tail risk measure derived from the 25-delta put/call spread

Derived features will include the absolute VRP (VIX minus forward realised vol), the relative VRP ($rVRP = (VIX - RV) / VIX$), and the term structure slope (VIX3M minus VIX9D). The rVRP will be winsorised at the 1st and 99th percentiles to limit the effect of periods where RV briefly

exceeds VIX. A 5-day rolling average of VIX will serve as the denominator to reduce noise in the relative measure.

Time alignment:

VIX at close on day t is a 30-day forward-looking measure. Realised volatility must therefore be computed over the *subsequent* period, not the preceding one, to avoid look-ahead bias. For the weekly forecast task, VIX9D at Friday close on week t is aligned with realised vol over week $t+1$. For the monthly task, end-of-month VIX is aligned with realised vol over the following month. All features will be constructed following this convention and alignment will be verified explicitly before any model training begins.

The train/test split is pre-2020 for training, 2020 to 2025 for testing. The test period covers two materially different volatility regimes: the COVID crash of March 2020 and the 2022 rate hiking cycle. The model is expected to underperform relative to in-sample fit across this period; this is a feature of the experimental design that tests generalisation to unseen regimes rather than a limitation.

The pipeline will be structured to accept richer options datasets without architectural changes.

Input Encoding:

Daily signals will be encoded as spike trains and presented to the network as sequences of five timesteps per week, one per trading day, so that intra-week temporal structure is preserved. The output is decoded at the end of each weekly window. Monthly forecasts use the same daily-encoded input over approximately twenty timesteps.

The delta encoding threshold will be set per-series using an adaptive rule: the threshold for series i is a fixed multiple of the rolling 4-week standard deviation of that series. This tracks local signal volatility rather than applying a single global threshold, which would be too tight in calm regimes and too loose during stress. The multiplier will be treated as a tunable hyperparameter with a dedicated sensitivity analysis in Phase 2.

Four encoding strategies will be evaluated:

Encoding	Description	Notes
Rate	Spike rate proportional to signal value	Baseline encoding, discards temporal structure
Latency	Earlier spike = larger value	Spike deletion risk near zero
Population	Gaussian receptive fields across value range	Robust. Used in Mohan et al.
Delta / Event	Spike on significant change; adaptive threshold	Primary hypothesis of project. Aligns with SNN paradigm

As delta encoding is the primary hypothesis, encoding returns and volatility increments rather than levels. This approach is informed by the spike-based encoding of Gao et al. (2021) and Ezinwoke and Amos (2025), adapted from price spike prediction to volatility increment prediction.

Network Architecture and Training:

The architecture follows Nowotny et al. (2025): an input population layer, a single recurrent LIF hidden layer (256 to 1024 neurons), and a non-spiking leaky integrator output layer.

Timescale mapping:

The time constants in Nowotny et al. (2025) are calibrated for speech data at millisecond resolution. This project uses a different temporal mapping: one simulation timestep

corresponds to one trading day. A week is therefore five timesteps and a month approximately twenty. Heterogeneous τ_{mem} and τ_{syn} will be rescaled relative to this mapping — homogeneous baseline values of 3 timesteps (τ_{mem}) and 1 timestep (τ_{syn}) will be used, with gamma distribution parameters set accordingly. This ensures the neuron dynamics operate at a timescale appropriate to the financial data. In the financial context, short time constants capture spot volatility while longer ones encode term structure, providing implicit multi-horizon representation.

The regression loss adapts the loss shaping principle of Nowotny et al.:

$$L = (1/T) \int_0^T (V_{\text{out}}(t) - \sigma_{\text{target}})^2 \exp(-t/T) dt$$

The exponential weighting avoids temporal pressure that would drive spike deletion in hidden neurons. Silent neuron recovery (incrementing incoming weights when a neuron fires nothing for a full epoch) and a learning rate ease-in schedule will be applied from the start of training.

Spike raster monitoring will be built into the training loop from the first epoch. Rasters will be inspected at fixed intervals to detect silent neuron pathology early. If more than 10% of hidden neurons are silent at any checkpoint, training will be paused and the recovery mechanism reviewed before continuing. Most hyperparameters will be fixed at the Nowotny et al. defaults. A targeted search will be conducted over three parameters: the encoding threshold multiplier, the learning rate, and the regularisation spike rate target.

Both e-prop and EventProp will be evaluated and compared. Baselines are GARCH(1,1), HAR-RV, a comparable LSTM, and the SNN on price history only. Results will be reported separately for absolute VRP and relative VRP, and separately for weekly and monthly horizons. Evaluation metrics are MSE, MAE, directional accuracy, QLIKE loss, and the Mincer-Zarnowitz R-squared. Statistical significance of forecast differences will be assessed via Diebold-Mariano tests.

Timeline

Phase	Tasks	Weeks
1. Data & Baseline	Data acquisition; RV aggregation to weekly and monthly; VRP and rVRP construction with alignment verification; GARCH and HAR-RV baselines at both horizons; initial mGeNN pipeline setup and SNN regression with rate encoding	1 to 4
2. Encoding	Spike raster diagnostic tooling; all four encoding strategies; adaptive threshold sensitivity analysis; spike deletion analysis; loss shaping; training safeguards	5 to 8
3.Options Integration	Options feature pipeline; delta encoding of vol increment signals; ablation study per input signal; rVRP winsorisation and smoothing validation	9 to 12
4. Architecture	Timescale rescaling; heterogeneous τ ; delay line inputs; e-prop vs EventProp comparison; targeted hyperparameter search over threshold multiplier, learning rate, and spike rate target	13 to 16
5. Write-up	Full test evaluation at weekly and monthly horizons; horizon degradation comparison across SNN and baselines; statistical tests; dissertation. Note: hyperparameter search in Phase 4 carries schedule risk; if it runs long, delay line	17 to 20

Phase	Tasks	Weeks
	inputs will be deferred to reduce scope rather than compress the write-up.	

Contribution to Knowledge

1. No published work targets the VRP or realised volatility with spiking architectures. This project provides the first benchmark of SNN-based volatility forecasting against GARCH and HAR-RV and establishes whether neuromorphic approaches are viable for this task at weekly and monthly horizons.
2. Delta/event encoding for volatility signals. Existing SNN finance work uses rate or population encoding. Encoding volatility increments as spike events with adaptive per-series thresholds aligns the input representation with the SNN's computational paradigm. This extends the spike encoding framework of Gao et al. (2021) and Ezinwoke and Amos (2025) from price spike prediction to volatility forecasting.
3. Options-derived SNN inputs. No published work uses CBOE-sourced implied volatility signals as SNN input features. The project quantifies the forecast improvement attributable to forward-looking options information over price history alone.
4. Relative VRP as a regression target. The project introduces and evaluates the rVRP alongside the absolute VRP, assessing whether level-normalised forecasts generalise better across volatility regimes. The comparison of absolute and relative VRP predictability at weekly and monthly horizons is itself a novel result.
5. Horizon degradation characterisation. The project produces a systematic comparison of how forecast accuracy changes with horizon for the SNN, HAR-RV, and LSTM baselines. This result has not been produced for SNN-based models and provides a reference point for future work.
6. Loss shaping for SNN regression. The loss shaping principle of Nowotny et al. (2025) was developed for classification. This project extends it to continuous regression, providing practical guidance for SNN applications requiring real-valued output.

Future Extensions

The following are natural next steps once the core forecasting objective is established.

Confidence-Filtered Forecasting

Following Allena (2025), the model could produce uncertainty estimates alongside point forecasts, acting only on high-confidence predictions. SNN-specific approaches include Monte Carlo dropout and exploiting the natural stochasticity of spike threshold dynamics. The latter has no equivalent in conventional uncertainty estimation.

Event-Conditional Forecasting

Forecasting volatility conditional on specific scheduled macro events, using prediction market probabilities as additional inputs alongside the options-derived signals. This would leverage the event-driven architecture more directly and targets the periods that drive the majority of realised volatility.

Shortfall-Aware Hedging Integration

The SNN volatility forecast and its uncertainty estimate would serve as state inputs to a shortfall-aware RL hedging agent following Hu et al. (2026). The SNN provides the forward-looking volatility signal; the RL agent translates it into hedge ratios that minimise shortfall probability under transaction costs. This would address the full pipeline from volatility signal to hedging decision that neither the forecasting nor the hedging literature currently provides.

Neuromorphic Hardware Deployment

Benchmark the trained model on Intel Loihi or IBM TrueNorth against a GPU baseline, measuring power consumption and latency for continuous streaming inference.

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Appendix A: CBOE Data Sources

Signal	URL	Available From
VIX daily	cboe.com/tradable_products/vix/vix_historical_data	1990
VIX9D, VIX3M, VIX6M	cboe.com/products/vix-index-volatility/volatility-indexes	2011
VVIX	cboe.com/tradable_products/vvix/vvix_historical_data	2006
SKEW Index	cboe.com/products/vix-index-volatility/volatility-indexes	1990